
Group 09

**PageRhythm
Automated Testing Document**

Version 1.0

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Revision History

Date	Version	Description	Author
12/29/2024	1.0	The initial version of the document	Nguyen Hoang Minh Luu Quoc Bao Trinh Nguyen Thao Vi

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1 Introduction

This document is intended to describe how automated testing was used to test PageRhythm, the application of our group. It lists the details of each test case, including test case name, test scripts, and test result as well as the set up environment when the test automation tool was used.

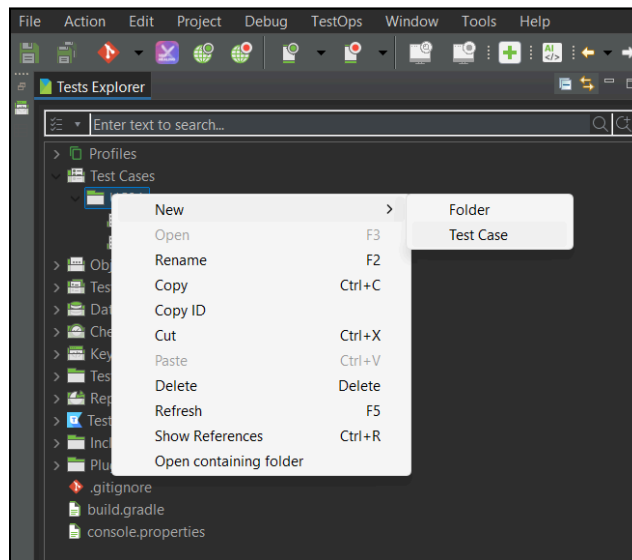
2 Used test automation tool

2.1 Tool

Our group used Katalon Studio to perform automated testing with our application, which is a web-based application.

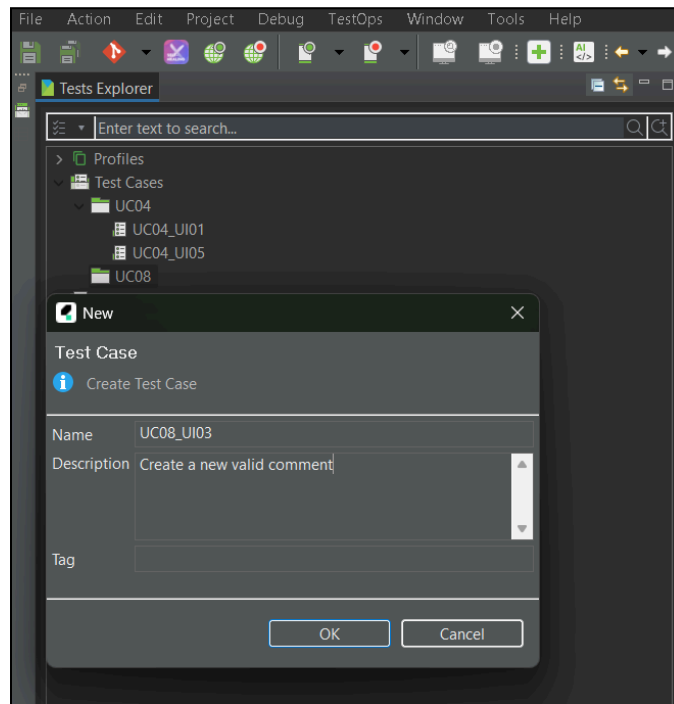
2.2 Process for creating scripts

This subsection outlines the process of creating test scripts for our test cases using Katalon Studio. The first step is to create a "Test Case", as illustrated in the image below.

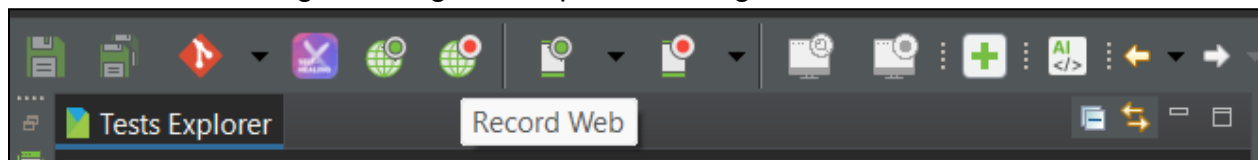


When creating a new test case in Katalon Studio's window, we enter the name in the format "<FeatureID>_<CaseID>" for the test case, and use the corresponding test case name from the "Test Cases.xlsx" file as the description. After filling in the required fields, we click the "Save" button to save the test case.

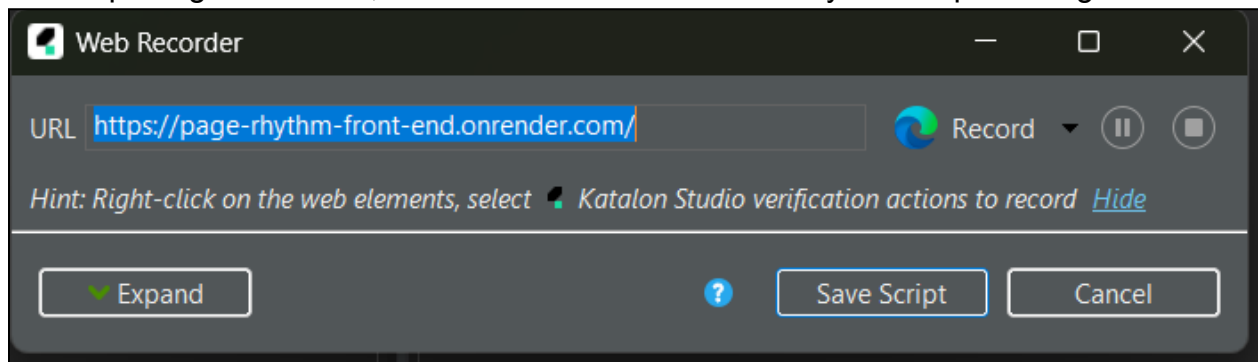
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Following the successful creation of the new test case, we used the “**Record web**” feature to start generating the scripts for testing our test cases.

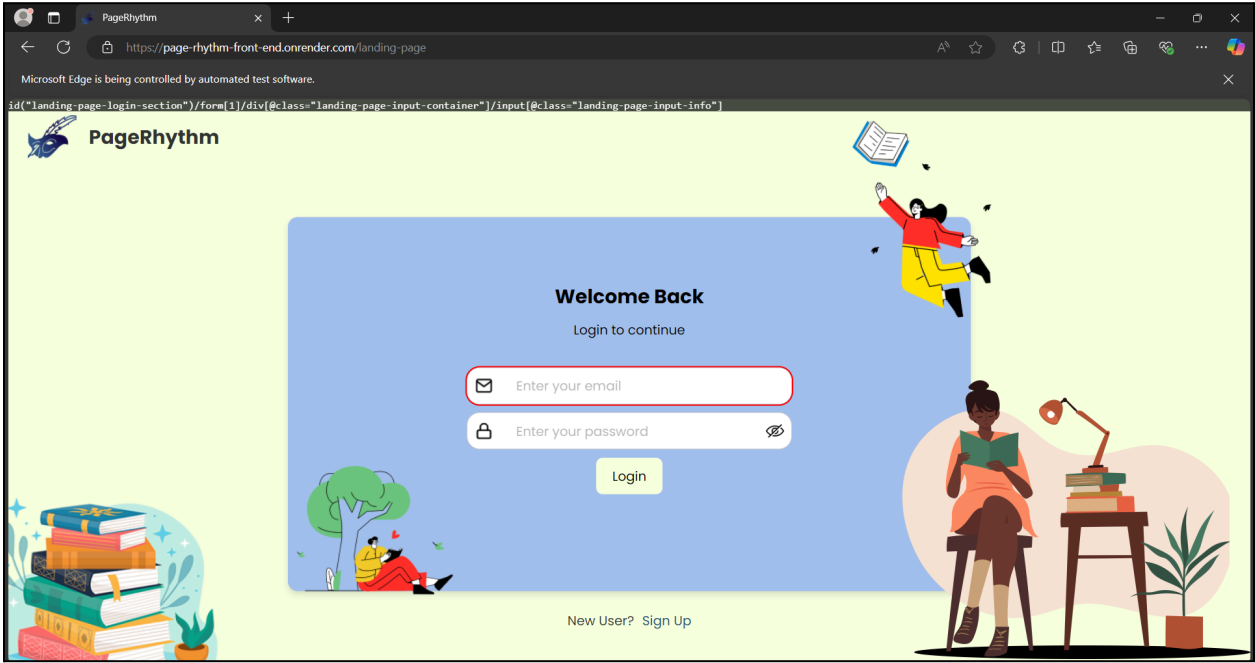


For recording our website, we entered the URL of our deployed application and chose “**Edge Chromium**” as the browser. Then, we clicked the “**Record**” button to start capturing the actions, which will later be automated by the scripts during execution.



A browser window will open, allowing us to perform the necessary actions that will be recorded and converted into automated scripts for future execution.

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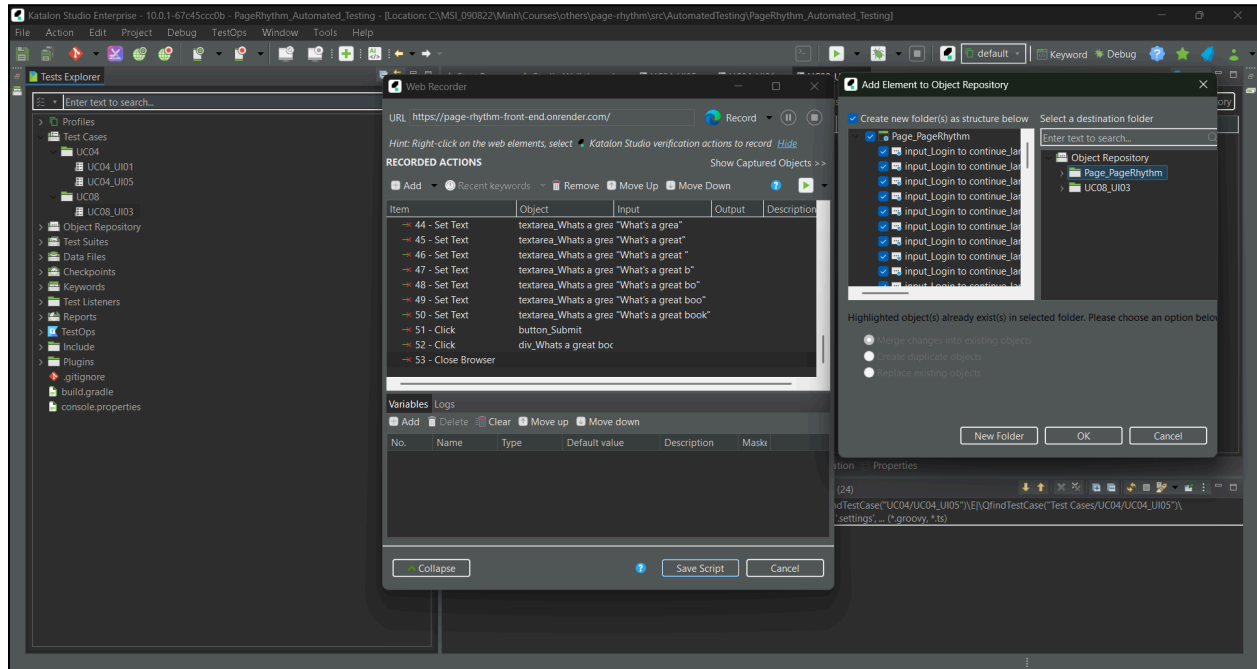


Specifically, for each test case, the actions were done according to the corresponding cell in the “**Test step**” column in the “**Test Cases.xlsx**” file.

49	Function 08: Rate and comments about books						
50	UC08	UI01	Add rating	1. On the page that shows the details of the books, click an empty star or an filled star that is not rightmost	1. The chosen star and the stars left to it (if any) are filled 2. The stars right to chosen star (if any) are empty	The same ER	Pass
51	UC08	UI02	Remove rating	1. On the page that shows the details of the books, click the rightmost filled star	1. All stars are empty	The same ER	Pass
52	UC08	UI03	Create a new valid comment	1. On the page that shows the details of the books, click the "Write a comment" button 2. Enter "What's a great book" into the box 3. Click the "Submit" button	1. A new comment with content: "What's a great book" appears	The same ER	Pass

After completing all the necessary steps to perform the test cases, we stopped the recording process and clicked the “**Save Script**” button. This action generates the automated script, which can be used for testing the functionality in future executions.

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To ensure the test scripts function as expected, we made adjustments and fine-tuned the scripts. This involved refining specific actions, adding necessary statements, and addressing any issues that could impact the accuracy and reliability of the automated tests.

These scripts were used to perform automated testing.

3 Test Cases, Test Scripts

3.1 Test cases

The below table lists the details of each test case on which the automated test was used, including the test case name, the result, the testers who write the test scripts and run these scripts, and its function ID and case ID.

The associated test script of each test case is placed in the [3.2 Test scripts](#) section.

In summary, the automated testing successfully executed all the test cases, with each one showing a "Passed" status.

Function/ Feature ID	Case ID	Test case name	Test result	Tester	Tested Date
UC01	UI01	Register with full of required information	Passed	Luu Quoc Bao	29/12/2024
UC01	UI03	Register with incorrect email format and full of other required information	Passed	Luu Quoc Bao	29/12/2024

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UC04	UI01	Move to next page for the book of more than one page	Passed	Nguyen Hoang Minh	29/12/2024
UC08	UI03	Create a new valid comment	Passed	Nguyen Hoang Minh	29/12/2024
UC08	UI04	Reply to a comment	Passed	Nguyen Hoang Minh	29/12/2024

3.2 Test scripts

3.2.1 UC01 - UI01

The test script for the test case UC01 - UI01 ("Register with full of required information" of "Function 08: Register an account")

```
import static com.kms.katalon.core.checkpoint.CheckpointFactory.findCheckpoint
import static com.kms.katalon.core.testcase.TestCaseFactory.findTestCase
import static com.kms.katalon.core.testdata.TestDataFactory.findTestData
import static com.kms.katalon.core.testobject.ObjectRepository.findTestObject
import static com.kms.katalon.core.testobject.ObjectRepository.findWindowsObject
import com.kms.katalon.core.checkpoint.Checkpoint as Checkpoint
import com.kms.katalon.core.cucumber.keyword.CucumberBuiltinKeywords as CucumberKW
import com.kms.katalon.core.mobile.keyword.MobileBuiltInKeywords as Mobile
import com.kms.katalon.core.model.FailureHandling as FailureHandling
import com.kms.katalon.core.testcase.TestCase as TestCase
import com.kms.katalon.core.testdata.TestData as TestData
import com.kms.katalon.core.testng.keyword.TestNGBuiltinKeywords as TestNGKW
import com.kms.katalon.core.testobject.TestObject as TestObject
import com.kms.katalon.core.webservice.keyword.WSBuiltInKeywords as WS
import com.kms.katalon.core.webui.keyword.WebUiBuiltInKeywords as WebUI
import com.kms.katalon.core.windows.keyword.WindowsBuiltinKeywords as Windows
import internal.GlobalVariable as GlobalVariable
import org.openqa.selenium.Keys as Keys
import java.util.Random

import com.kms.katalon.core.webui.keyword.WebUiBuiltInKeywords as WebUI
import com.kms.katalon.core.testobject.ObjectRepository as ObjectRepository

Random random = new Random()
int randomNumber = random.nextInt(1000)
```


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```

String username = "user${randomNumber}"
String email = "user${randomNumber}@test.com"
String password = "Pass@${randomNumber}"

WebUI.openBrowser('')

WebUI.navigateToUrl('https://page-rhythm-front-end.onrender.com')

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/a_Sign Up'))

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'))

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'),
    username)

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'),
    '')

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'),
    email)

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-fir_7c751d'),
    password)

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-sec_cbf5a7'),
    password)

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/Date input'),
    '22-12-2222')

```

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```

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/textarea_d'), 'f')

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/input_I agree with
the_agree-terms'))

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/button_Sign up'))

WebUI.waitForElementVisible(findTestObject('Object Repository/Page_PageRhythm/a_My
Library'), 10)

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/a_My Library'))

WebUI.closeBrowser()

```

3.2.2 UC01 - UI03

The test script for the test case UC01 - UI03 (“Register with incorrect email format and full of other required information” of “Function 08: Register an account”)

```

import static com.kms.katalon.core.checkpoint.CheckpointFactory.findCheckpoint
import static com.kms.katalon.core.testcase.TestCaseFactory.findTestCase
import static com.kms.katalon.core.testdata.TestDataFactory.findTestData
import static com.kms.katalon.core.testobject.ObjectRepository.findTestObject
import static com.kms.katalon.core.testobject.ObjectRepository.findWindowsObject
import com.kms.katalon.core.checkpoint.Checkpoint as Checkpoint
import com.kms.katalon.core.cucumber.keyword.CucumberBuiltinKeywords as CucumberKW
import com.kms.katalon.core.mobile.keyword.MobileBuiltInKeywords as Mobile
import com.kms.katalon.core.model.FailureHandling as FailureHandling
import com.kms.katalon.core.testcase.TestCase as TestCase
import com.kms.katalon.core.testdata.TestData as TestData
import com.kms.katalon.core.testng.keyword.TestNGBuiltinKeywords as TestNGKW
import com.kms.katalon.core.testobject.TestObject as TestObject
import com.kms.katalon.core.webservice.keyword.WSBuiltInKeywords as WS
import com.kms.katalon.core.webui.keyword.WebUiBuiltInKeywords as WebUI
import com.kms.katalon.core.windows.keyword.WindowsBuiltinKeywords as Windows
import internal.GlobalVariable as GlobalVariable
import org.openqa.selenium.Keys as Keys

```

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```

import java.util.Random
import com.kms.katalon.core.webui.keyword.WebUiBuiltInKeywords as WebUI
import com.kms.katalon.core.testobject.ObjectRepository as ObjectRepository

Random random = new Random()
int randomNumber = random.nextInt(1000)

String username = "user${randomNumber}"
String email = "user${randomNumber}@test.com"
String password = "Pass@${randomNumber}"

WebUI.openBrowser('')

WebUI.navigateToUrl('https://page-rhythm-front-end.onrender.com')

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/a_Sign Up'))

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'))

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'),
username)

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'),
'')

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-input-info'),
email)

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-fir_7c751d'),
password)

```

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```

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/input_Sign up to
continue_register-page-sec_cbf5a7'),
    password)

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/Date input'),
    '22-12-2222')

WebUI.setText(findTestObject('Object Repository/Page_PageRhythm/textarea_d'), 'f')

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/input_I agree with
the_agree-terms'))

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/button_Sign up'))

WebUI.waitForElementVisible(findTestObject('Object Repository/Page_PageRhythm/a_My
Library'), 10)

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/a_My Library'))

WebUI.closeBrowser()

```

3.2.3 UC04 - UI01

The test script for the test case UC04 - UI01 (“Move to next page for the book of more than one page” of “Function 04: Read and navigate books”)

```

import static com.kms.katalon.core.checkpoint.CheckpointFactory.findCheckpoint
import static com.kms.katalon.core.testcase.TestCaseFactory.findTestCase
import static com.kms.katalon.core.testdata.TestDataFactory.findTestData
import static com.kms.katalon.core.testobject.ObjectRepository.findTestObject
import static com.kms.katalon.core.testobject.ObjectRepository.findWindowsObject
import com.kms.katalon.core.checkpoint.Checkpoint as Checkpoint
import com.kms.katalon.core.cucumber.keyword.CucumberBuiltinKeywords as CucumberKW
import com.kms.katalon.core.mobile.keyword.MobileBuiltInKeywords as Mobile
import com.kms.katalon.core.model.FailureHandling as FailureHandling
import com.kms.katalon.core.testcase.TestCase as TestCase
import com.kms.katalon.core.testdata.TestData as TestData
import com.kms.katalon.core.testng.keyword.TestNGBuiltinKeywords as TestNGKW
import com.kms.katalon.core.testobject.TestObject as TestObject

```

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```

import com.kms.katalon.core.webservice.keyword.WSBuiltInKeywords as WS
import com.kms.katalon.core.webui.keyword.WebUiBuiltInKeywords as WebUI
import com.kms.katalon.core.windows.keyword.WindowsBuiltInKeywords as Windows
import internal.GlobalVariable as GlobalVariable
import org.openqa.selenium.Keys as Keys

WebUI.openBrowser('')

WebUI.navigateToUrl('https://page-rhythm-front-end.onrender.com/')

String email = 'nguyenvanmuoi@gmail.com'
WebUI.setText(findTestObject('Object Repository/UC08_UI04/Page_PageRhythm/input_Login
to continue_landing-page-input-info'), email)

WebUI.setEncryptedText(findTestObject('Object Repository/Page_PageRhythm/input_Login
to continue_password-input'), '8SQVv/p9jVTqW8DV17zZxg==')

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/button_Login'))

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/img'))

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/button_Read This
Book'))

WebUI.click(findTestObject('Object Repository/Page_PageRhythm/img_Page 1 2 of
3_read-book-page-navigation-icon'))

WebUI.closeBrowser()

```

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3.2.4 UC08 - UI03

The test script for the test case UC08 - UI03 ("Create a new valid comment" of "Function 08: Rate and comments about books")

```
import static com.kms.katalon.core.checkpoint.CheckpointFactory.findCheckpoint
import static com.kms.katalon.core.testcase.TestCaseFactory.findTestCase
import static com.kms.katalon.core.testdata.TestDataFactory.findTestData
import static com.kms.katalon.core.testobject.ObjectRepository.findTestObject
import static com.kms.katalon.core.testobject.ObjectRepository.findWindowsObject
import com.kms.katalon.core.checkpoint.Checkpoint as Checkpoint
import com.kms.katalon.core.cucumber.keyword.CucumberBuiltinKeywords as CucumberKW
import com.kms.katalon.core.mobile.keyword.MobileBuiltInKeywords as Mobile
import com.kms.katalon.core.model.FailureHandling as FailureHandling
import com.kms.katalon.core.testcase.TestCase as TestCase
import com.kms.katalon.core.testdata.TestData as TestData
import com.kms.katalon.core.testng.keyword.TestNGBuiltinKeywords as TestNGKW
import com.kms.katalon.core.testobject.TestObject as TestObject
import com.kms.katalon.core.webservice.keyword.WSBuiltInKeywords as WS
import com.kms.katalon.core.webui.keyword.WebUiBuiltInKeywords as WebUI
import com.kms.katalon.core.windows.keyword.WindowsBuiltinKeywords as Windows
import internal.GlobalVariable as GlobalVariable
import org.openqa.selenium.Keys as Keys

WebUI.openBrowser('')

WebUI.navigateToUrl('https://page-rhythm-front-end.onrender.com/landing-page')

String email = 'nguyenvanmuoi@gmail.com'
WebUI.setText(findTestObject('Object Repository/UC08_UI04/Page_PageRhythm/input_Login
to continue_landing-page-input-info'), email)

WebUI.setEncryptedText(findTestObject('Object
Repository/UC04_UI01/Page_PageRhythm/input_Login to continue_password-input'),
    '8SQVv/p9jVTqW8DV17zZxg==')
```

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```

WebUI.click(findTestObject('Object
Repository/UC04_UI01/Page_PageRhythm/button_Login'))

WebUI.click(findTestObject('Object Repository/UC04_UI01/Page_PageRhythm/h3_Learn
Flask I'))

WebUI.click(findTestObject('Object Repository/UC04_UI01/Page_PageRhythm/button_Write
a comment'))

String comment = 'What\'s a great book'
WebUI.setText(findTestObject('Object
Repository/UC04_UI01/Page_PageRhythm/textarea_Whats a great book'), comment)

WebUI.click(findTestObject('Object
Repository/UC04_UI01/Page_PageRhythm/button_Submit'))

WebUI.click(findTestObject('Object Repository/UC04_UI01/Page_PageRhythm/div_Whats a
great book'))

WebUI.closeBrowser()

```

3.2.5 UC08 - UI04

The test script for the test case UC08 - UI04 (“Reply to a comment” of “Function 08: Rate and comments about books”)

```

import static com.kms.katalon.core.checkpoint.CheckpointFactory.findCheckpoint
import static com.kms.katalon.core.testcase.TestCaseFactory.findTestCase
import static com.kms.katalon.core.testdata.TestDataFactory.findTestData
import static com.kms.katalon.core.testobject.ObjectRepository.findTestObject
import static com.kms.katalon.core.testobject.ObjectRepository.findWindowsObject
import com.kms.katalon.core.checkpoint.Checkpoint as Checkpoint
import com.kms.katalon.core.cucumber.keyword.CucumberBuiltinKeywords as CucumberKW
import com.kms.katalon.core.mobile.keyword.MobileBuiltInKeywords as Mobile
import com.kms.katalon.core.model.FailureHandling as FailureHandling
import com.kms.katalon.core.testcase.TestCase as TestCase

```

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```

import com.kms.katalon.core.testdata.TestData as TestData
import com.kms.katalon.core.testng.keyword.TestNGBuiltinKeywords as TestNGKW
import com.kms.katalon.core.testobject.TestObject as TestObject
import com.kms.katalon.core.webservice.keyword.WSBuiltInKeywords as WS
import com.kms.katalon.core.webui.keyword.WebUiBuiltInKeywords as WebUI
import com.kms.katalon.core.windows.keyword.WindowsBuiltinKeywords as Windows
import internal.GlobalVariable as GlobalVariable
import org.openqa.selenium.Keys as Keys

WebUI.openBrowser('')

WebUI.navigateToUrl('https://page-rhythm-front-end.onrender.com/landing-page')

String email = 'nguyenvanmuoi@gmail.com'
WebUI.setText(findTestObject('Object Repository/UC08_UI04/Page_PageRhythm/input_Login
to continue_landing-page-input-info'), email)

WebUI.setEncryptedText(findTestObject('Object
Repository/UC08_UI04/Page_PageRhythm/input_Login to continue_password-input'),
    '8SQVv/p9jVTqW8DV17zZxg==')

WebUI.click(findTestObject('Object
Repository/UC08_UI04/Page_PageRhythm/button_Login'))

WebUI.click(findTestObject('Object Repository/UC08_UI04/Page_PageRhythm/h3_Learn
Flask I'))

WebUI.click(findTestObject('Object Repository/UC08_UI04/Page_PageRhythm/button_Reply
comment'))

String comment = 'Totally agree'
WebUI.setText(findTestObject('Object
Repository/UC08_UI04/Page_PageRhythm/textarea_TotalComment'), comment)

WebUI.click(findTestObject('Object
Repository/UC08_UI04/Page_PageRhythm/button_Submit'))

```


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```
WebUI.closeBrowser()
```